

News & Social Media Analytics Engine
Data Structures and Algorithms -3 (25CS2103E)

in

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Submitted by
D Nikhil Krishna (2520090025)
B Purna (2520090188)
V Adithya (2520080036)

Team Number: 17
Section: 13

Under the guidance of

Dr Dyuti Banerjee
Asst. Prof ,AIDS



Hyderabad-500090, Telangana, India.

ABSTRACT

The News & Social Media Analytics Engine is a DSA-based system designed to efficiently process and organize large volumes of news articles and social media posts. The project applies fundamental data structures and algorithms to solve practical problems in media analysis.

Hash tables are used for fast duplicate-content detection, queues manage incoming posts, heaps prioritize trending or important stories, and graphs represent relationships between related news items. Trees and sorting algorithms support efficient searching, classification, and ranking of content.

The system can group related stories, identify duplicate posts, rank trending content, and assign high-priority posts for moderation. By applying efficient searching, hashing, sorting, graph traversal, and priority-based processing techniques, the system demonstrates how DSA concepts can be used to handle large datasets efficiently.

The project aims to provide a structured and scalable approach to news and social media content analysis.

Introduction

The rapid growth of online news and social media platforms produces a huge amount of content every day. Processing this information manually is difficult because many articles and posts may contain duplicate or related information. Efficient algorithms and data structures are therefore required to organize, search, compare, and prioritize this content.

The News & Social Media Analytics Engine applies Data Structures and Algorithms to these problems. Incoming posts can be maintained using queues, while hash tables provide fast detection of duplicate content. Graphs can represent relationships between related stories, and heaps or priority queues can be used to identify and prioritize trending or important posts. Sorting and searching algorithms help rank and retrieve information efficiently.

The project demonstrates how DSA concepts can be combined to build an efficient system capable of processing and organizing large quantities of media data.

Software and hardware requirements

Software Requirements

- Programming Language: Python
- IDE: VS Code / PyCharm
- Operating System: Windows or Linux
- Database: MySQL / SQLite
- Libraries: Standard Python libraries and required data-processing libraries
- Version Control: Git

Hardware Requirements

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB or above
- Storage: 256 GB SSD or above
- Network: Internet connection for collecting news/social media data

Signature of the Guide

Course Instructor